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Maharaja Institute of Technology Mysore

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Acknowledgement

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Thank you.

MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE

(Affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING & DATA SCIENCE

Declaration

This is to certify that NSS Activity points submitted by **Ananya S** bearing the USN **4MH24EC009** is the students of MAHARAJA INSTITUTE OF TECHNOLOGY MYSORE and this report satisfies the academic requirements in partial fulfilment for the award of the degree of Bachelor of Engineering and approved to carry out for academic year 2024-2025

Signature of NSS

Co Ordinator

Signature of HoD

Dept. of CSE(AI)

Signature of Principal

FOOD PRESERVATION/ PACKING ACTIVITY

Conducted at: Independence Company, 33/B, Mysuru, Ilavala Hobli, Karnataka 570018

Duration: 3 Days

Date: 05/07/2025 to 07/07/2025

INTRODUCTION

The Story Behind Every Biscuit

Every time we enjoy a crispy biscuit with tea or coffee, we rarely think about the journey it took to reach us—starting from the raw ingredients like wheat flour and sugar, moving through mixing, baking, cooling, quality checking, packaging, and finally being transported to supermarkets. Behind the scenes, there lies a carefully maintained balance of **preservation techniques and smart packaging technologies** that ensure the biscuit remains fresh, crunchy, and safe until it is consumed.

The food processing industry, especially the **biscuit sector**, relies heavily on **scientific food preservation** and **packaging systems**. These systems not only improve shelf life and protect the product but also communicate essential information such as ingredients, nutritional values, allergens, expiry dates, and branding. In short, preservation and packaging are what transform a simple baked product into a market-ready, customer-safe experience.

Why Preservation and Packaging Matter in Biscuit Manufacturing

Biscuits are dry, baked food products that have relatively low moisture content. However, they are **extremely sensitive to moisture, air, and microbial contamination**, which can lead to sogginess, loss of crispness, or even spoilage if not preserved properly. Without appropriate techniques:

- **Air exposure** causes oxidation, making the biscuits lose aroma and flavor.
- **Moisture** causes sogginess and microbial growth.
- **Physical damage** during transport can reduce shelf appeal.
- **Sunlight and heat** can break down fats/oils, leading to rancidity.

Therefore, to preserve taste, texture, and shelf life, the biscuit industry must adopt reliable **preservation strategies** (like humidity control, nitrogen flushing, and sealing) and combine them with **intelligent packaging systems** (like vacuum pouches, foil laminates, or PET trays).

Overview of the Program

To create awareness and provide **first-hand industry experience**, a 3-day educational visit and training activity was conducted at **Independence**, a Mysuru-based biscuit manufacturing company located in Ilavala Hobli, Karnataka. The event allowed participants—mostly students and food science enthusiasts—to explore every step of how biscuits are manufactured, preserved, and packed for the market.

The program offered:

- A guided tour of the **biscuit production line**
- Exposure to **preservation techniques used post-baking**
- Practical learning on **automated packaging systems**

- Safety protocols, hygiene practices, and quality control measures
- Insights into modern **branding and labeling methods**

This activity was not just informative but also deeply inspiring, especially for students pursuing food technology, industrial engineering, biotechnology, and business studies. By being on the production floor and observing real-time processes, they learned how various disciplines come together to create products they see in stores every day.

The Scientific Side of a Biscuit

Although a biscuit seems simple on the surface, it is actually a product of careful formulation and science. Each ingredient contributes to shelf life, texture, and taste. For example:

- **Fat** gives richness but also attracts oxygen (which causes rancidity).
- **Sugar** attracts moisture, so storage must be dry.
- **Flour** needs to be stored safely before use, free from pests.

Once biscuits are baked, they pass through **cooling tunnels**, after which they must be **immediately preserved** in an environment that prevents exposure to:

- Air (oxygen)
- Humidity
- Dust and microbes

These are typically managed through:

- Nitrogen flushing (displaces oxygen)
- Vacuum sealing
- Multilayer barrier films (laminates)
- Desiccant sachets inside bulk packs

Students got to witness these practices not just in theory, but directly as the biscuits moved from the **oven to conveyor belts to packaging units**.

Packaging: More Than Just a Wrapper

Participants learned how packaging is not just about wrapping the biscuit; it is a **multifunctional process** that involves:

- **Protection:** Keeping biscuits from moisture, dust, breakage
- **Communication:** Displaying nutritional values, barcode, brand identity
- **Convenience:** Easy-to-open designs, resealable packs, portion control
- **Marketing:** Appealing colors, logos, mascots, and catchy taglines

The factory tour revealed various **packaging materials** used by the company:

- Aluminum foil wraps for moisture barrier
 - Laminated plastic films for multi-layer protection
 - Carton boxes with dividers for bulk packaging
 - Printed wrappers with batch codes and expiry dates
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Sustainability in Biscuit Packaging

A highlight of the visit was how **Independence Biscuit Company** is incorporating **eco-friendly alternatives** to reduce plastic use.

They are exploring:

- **Compostable packaging**
- **Paper-based trays**
- **Biodegradable films**
- **Bulk carton refills** for institutional use (like hostels, schools, etc.)

Students were encouraged to brainstorm packaging designs that combine **efficiency, affordability, and eco-consciousness**. This also included discussions on **Extended Producer Responsibility (EPR)** and how manufacturers must consider packaging waste management.

Bridging Academia and Industry

One of the biggest takeaways was how the event helped **bridge the gap between classroom theory and industrial reality**. Students had learned about food preservation, shelf-life testing, packaging standards (like FSSAI and ISO), and cold chains in class. But this visit helped them:

- See actual baking ovens and packaging machines in action
- Understand why humidity and temperature are monitored in real-time
- Watch how packaging errors are flagged and corrected
- Learn from industry experts how packaging influences **customer trust**

Many students expressed that this event changed how they look at packaged food, and it gave them the motivation to innovate in the future—whether as food technologists, packaging designers, or startup founders.

Final Words on the Introduction

This report begins by acknowledging that **a simple biscuit** holds within it the power of **science, safety, and sustainability**. The 3-day event organized at **Independence Biscuit Company** was more than a visit—it was an experience that opened minds, inspired futures, and celebrated the unseen work that goes into every crunchy bite.

Through this report, we aim to capture that experience in its full richness, starting with the background and objectives, leading into each day's activities, and concluding with the lessons that students will carry forward into their careers.

OBJECTIVE OF THE ACTIVITY

Setting the Purpose

The main intention behind organizing this 3-day food preservation and packaging activity at **Independence Biscuit Company**, Mysuru, was to go beyond textbooks and allow students to witness, participate in, and understand how the **real food industry operates**, especially in the **biscuit production sector**.

This was not just a study visit—it was a carefully structured **experiential learning program**. The activity was designed with the idea that students learn better when they see, touch, and try what they've only read about.

Let's explore the **five core objectives** of the program in detail:

Objective 1: Understanding Food Preservation in Biscuit Manufacturing

Preservation is the heart of the food industry. The objective here was to help students understand:

- **What causes biscuits to spoil**, go soft, or lose flavor over time.
- How elements like **air, moisture, and temperature** affect dry baked goods.
- What natural and artificial preservation methods are applied in biscuit factories.

In particular, they were introduced to methods like:

- **Nitrogen flushing:** Replacing oxygen with nitrogen in sealed packs to prevent oxidation.
- **Humidity control** in the baking and cooling zones.
- Use of **oxygen absorbers and desiccants** in larger boxes.

Students got to observe the use of **moisture meters, air-sealed machines, and automated tunnel coolers**—tools that ensure the biscuits are preserved immediately after baking.

By the end of this segment, students were able to **differentiate preservation strategies** used for biscuits (dry foods) vs. dairy, frozen, or liquid foods.

Objective 2: Learning the Role of Packaging in Shelf Life and Branding

Packaging is not only the final step of production but also a critical factor in product success. This objective was to help participants:

- See how biscuits are **sorted, wrapped, sealed, and boxed** using high-speed machinery.
- Learn about the **layers of materials** used: foil for moisture resistance, plastic for protection, and cartons for stacking.
- Understand how the **design of a biscuit wrapper** reflects the brand, audience, and usage.

Students examined packaging types like:

- **Single-serve pouches** for retail counters.
- **Multi-pack wrappers** with tear-notches and resealable options.
- **Bulk corrugated boxes** for wholesale and institutional supply.

Through live demonstrations, they saw the **precise timing and calibration** needed in automated packaging lines. A key highlight was understanding how **wrong temperature, misaligned wrappers, or air leakage** can lead to spoilage or customer rejection.

This gave students deep insights into the **technical and creative** aspects of packaging:

- Barcoding and batch printing
 - Date stamping and FSSAI labeling
 - Use of child-safe or easy-open designs
-

Objective 3: Exposure to Real-Time Industry Practices

While colleges teach theory, industry shows the **practical constraints and efficiencies**. This objective focused on bridging that gap:

- Students observed how production shifts are timed.
- They witnessed how **downtime is minimized** during changeovers between biscuit variants (e.g., cream vs. plain).
- They learned how hygiene is maintained via **gloves, airlocks, and uniform codes**.

The students also interacted with:

- **Machine operators** who explained how packing speed is controlled
- **Quality analysts** who described how defective biscuits are rejected
- **Maintenance staff** who monitor machine health and performance

This exposure helped them appreciate the **teamwork, discipline, and technical precision** required in food industries. Many students realized that **each step is interdependent**—a delay in sealing can ruin the whole batch, or a labeling mistake can lead to a legal problem.

Objective 4: Promoting Sustainability and Innovation in Packaging

Another major goal of the activity was to spark creative thinking about how food industries can **reduce their environmental footprint** while still delivering quality.

Students were introduced to concepts like:

- **Biodegradable wrappers** made from starch films or paper laminates
- **Recyclable packaging formats**
- **Minimalist design** that uses less ink and fewer layers

They also discussed **green logistics**—using box sizes that reduce wasted space in transport, leading to fuel efficiency and cost savings.

This objective was especially important because the next generation of engineers and designers must consider:

- **What happens to packaging after it's discarded?**
- **Can we reduce plastics without compromising shelf life?**
- **How can packaging inspire trust and loyalty while being eco-friendly?**

Participants were encouraged to brainstorm their own packaging solutions, keeping in mind **cost, functionality, safety, and sustainability**.

Objective 5: Encouraging Career Exploration and Industry Readiness

Many students are unaware of the wide range of career paths available in food-related industries. This activity helped them explore:

- Roles like **Food Technologist, Quality Control Analyst, Packaging Designer, Machine Operator, Product Development Scientist, and Supply Chain Manager.**
- How diverse departments (R&D, production, logistics, design, finance) contribute to one final biscuit.
- What skills are expected by industry: teamwork, attention to detail, hygiene understanding, and machine handling.

Speakers from the company shared their career stories and explained how a small idea—like reducing biscuit breakage in a packet—can lead to a huge cost saving or brand improvement.

Students were also introduced to certifications like:

- **FSSAI regulations**
- **ISO 22000 (Food Safety Management)**
- **HACCP (Hazard Analysis Critical Control Point)**

This opened their eyes to how **safe food production is not only a science but a regulated responsibility.**

Summary of Objectives

The 3-day activity at Independence Biscuit Company was structured around **holistic learning**. It wasn't limited to machinery or raw materials—it went deeper into understanding:

- The science of freshness
- The power of design
- The responsibility of safety
- The art of process efficiency
- The future of sustainable food systems

By the end of the event, students walked away not just with observations but with a **newfound respect for the biscuit industry**, ideas for innovation, and possibly, inspiration for their careers ahead.

ABOUT INDEPENDENCE BISCUIT COMPANY

“Arrival at Independence Biscuit Company, Mysuru – the facility entrance.”



A Homegrown Brand with Global Standards

Independence Biscuit Company, located in the serene and industrially active area of **Ilavala Hobli, Mysuru**, Karnataka, is a modern Indian success story in the field of **processed food manufacturing**. Though focused primarily on biscuit production, the company embodies a complete ecosystem of food safety, industrial precision, innovation, and environmental responsibility.

Its journey started with a simple vision: to bring **high-quality, hygienically processed, and affordably priced biscuits** to households across India. Today, it has grown into a full-scale biscuit processing and packaging unit with **automated lines, advanced lab facilities, and eco-conscious practices**—all while staying rooted in local values.

Product Focus: Biscuits for Every Generation

The company's flagship products include:

- **Classic glucose biscuits** – for energy and affordability
- **Cream-filled biscuits** – for children and gifting packs
- **Digestive and oat biscuits** – for health-conscious consumers
- **Butter and milk biscuits** – for tea-time snacks
- **Customized bulk packs** – for institutions like schools, hospitals, and hostels

Each product undergoes **formulation, baking, cooling, preservation, and packaging** in a controlled facility that maintains **hygiene, nutritional consistency, and crispness**—essential features for biscuits.

The company has established long-standing trust among **retailers, distributors, and end-consumers**, and continues to upgrade its infrastructure in line with national and global standards.

Factory Infrastructure and Departments

The Independence facility is spread across a strategically located industrial plot at **33/B, Ilavala Hobli, Mysuru**. It is divided into well-functioning departments, each responsible for a stage in the biscuit-making and packaging process:

1. Raw Material Storage & Sorting

- Ingredients like refined wheat flour, sugar, vegetable oil, milk solids, and raising agents are stored in temperature-controlled silos.
- Food-grade containers and pest-proof bins ensure hygiene.
- Inputs are passed through **metal detectors and sieves** before processing.

2. Mixing & Formulation Area

- Dough is prepared using high-capacity mixers based on exact recipes.
- Variants (e.g. milk, butter, or chocolate) are separated here.
- Measured water and emulsifiers are added for consistency.

3. Baking Section

- Conveyor belt ovens ensure **uniform baking** at precise temperatures.
- Each batch is monitored using sensors and real-time temperature logs.
- Biscuits are cooled on stainless-steel belts after baking.

4. Preservation and Cooling

- Special cooling tunnels bring biscuit temperature to ambient levels.

- This prevents **condensation**, which can lead to soggy or microbial growth.
- Some packaging units use **nitrogen flushing** to remove oxygen before sealing.

5. Packaging Lines

- Fully automated packaging machines wrap biscuits at high speed (hundreds per minute).
- Machines also handle **coding, labeling, batch tracking**, and **weigh-checking**.
- Final packs are sorted by size (e.g. ₹5, ₹10, ₹20 packs) and placed into cartons.

6. Quality Control Laboratory

- Samples from each batch are tested for:
 - **Moisture content**
 - **Crispness and breakage**
 - **Shelf-life behavior under different temperatures**
 - **Microbial and fungal load**
- A small **R&D team** works on new variants and improving texture, flavor, and packaging durability.

7. Dispatch & Logistics

- Ready cartons are stored in clean, dry dispatch rooms.
- Goods are sent to **distributors and wholesalers** via logistics partners.
- The dispatch team ensures that products are rotated using the **FIFO system** (First In, First Out) to maintain freshness.

Focus on Hygiene and Food Safety

The company proudly follows food industry guidelines such as:

- **FSSAI (Food Safety and Standards Authority of India)**
- **ISO 22000 (Food Safety Management System)**
- **HACCP (Hazard Analysis and Critical Control Points)**

Some of the hygiene protocols observed include:

- Compulsory use of gloves, aprons, caps, and shoe covers by staff
- Daily sanitation of workspaces using food-safe disinfectants
- Air curtains and **pest control systems** at all entry points
- Regular audits and surprise inspections to ensure compliance

Visitors to the factory—including students from this 3-day event—were required to undergo a safety briefing and wear clean room attire to prevent contamination.

Sustainability and Eco-Conscious Operations

Despite being a mid-sized unit, Independence Biscuit Company places a strong emphasis on **sustainable practices**, such as:

- Use of **biodegradable packaging wrappers** for specific product lines
- Energy-efficient ovens and LED lighting across the plant
- **Rainwater harvesting** and waste-water recycling within the campus
- Packaging designed to **reduce space wastage in transport**, improving fuel efficiency

The company has started phasing out multi-layered plastic in favor of:

- **Compostable pouches**
- **Recyclable outer cartons**

- **Mono-layer polyethylene films**, which are easier to recycle

It is also collaborating with local agencies for **waste recovery and plastic offsetting**, aligning with the principles of **Extended Producer Responsibility (EPR)**.

Learning Opportunities and Educational Involvement

The company has an open-door policy for academic collaboration. It regularly:

- Hosts **industrial visits** and training programs for college students
- Offers **internship opportunities** in food safety, QA, mechanical maintenance, and logistics
- Participates in local **career fairs and technical fests** to recruit young talent
- Donates surplus inventory to NGOs through properly packed and labeled consignments

During the 3-day event, students interacted with:

- **Production Managers** who explained daily schedules and output goals
- **Packaging Engineers** who showcased how machines are calibrated
- **Lab Technicians** who demonstrated microbial testing techniques
- **Design team members** who explained branding and compliance packaging

This experience helped students visualize **real-world applications** of their classroom concepts like automation, preservation, packaging science, and industrial safety.

Why This Company Was the Right Choice

Independence Biscuit Company was chosen for this educational activity for several strong reasons:

- It is a **local success story** that operates at national quality levels.
- The company works across all levels of biscuit production—**formulation, preservation, packaging, and dispatch**.
- Its facilities are advanced, but still relatable for students.
- The staff is **educator-friendly**, with patient, knowledgeable professionals who support learning.
- Its processes align with modern goals like **food safety, affordability, and sustainability**.

Conclusion: A Model for Food Industry Learning

The Independence Biscuit Company proved to be more than just a food production facility—it became a **classroom of living machines, dedicated professionals, and scientific insight**. For the students, this company was a real-world example of how:

- Science and manufacturing meet
- Preservation protects public health
- Packaging connects brand to consumer
- And how an Indian business can embrace innovation with responsibility

This section aims to document not only what the company does, but **how it operates with values**, and how it reflects the **future of Indian food technology**—responsible, efficient, and student-friendly.

IMPORTANCE OF FOOD PRESERVATION AND PACKAGING

Why Food Preservation and Packaging Matter

In today's fast-paced world, the way food is preserved and packaged plays a crucial role in ensuring **safety, quality, freshness, and consumer satisfaction**. It's no longer just about producing tasty food—it's about **delivering it to consumers in perfect condition**, without contamination, spoilage, or loss of nutritional value.

For biscuit manufacturing industries like **Independence Biscuit Company**, food preservation and packaging are **the backbone of production**. Biscuits, though considered shelf-stable, are highly sensitive to **moisture, air, and microbial spoilage**. A single packaging error or preservation lapse can lead to:

- Loss of crispness and texture
- Growth of mold or spoilage bacteria
- Unpleasant odor and taste due to oxidation
- Customer dissatisfaction and product returns

Thus, implementing the **right preservation techniques** and using the **appropriate packaging materials** is essential to:

- **Extend the shelf life** of the product
- **Maintain hygiene and safety**
- **Ensure market competitiveness**
- **Prevent food wastage and economic loss**

Preservation in the Biscuit Industry

Although biscuits are dry foods, they can easily absorb moisture from the air, especially in humid regions like South India. This can lead to sogginess or microbial growth. So, the **biscuit industry adopts several preservation strategies**, such as:

- **Cooling tunnels:** Reduce product temperature quickly post-baking to prevent condensation.

- **Dehumidifiers** in packing rooms: Keep air dry to avoid moisture contact.
- **Nitrogen flushing:** Replace oxygen inside the packaging with nitrogen to prevent oxidation.
- **Sealed wrapping:** Prevent air and microbial entry.

These steps ensure that biscuits stay **crisp, safe, and flavorful** for months after production—even in harsh climates.

Packaging: The First Line of Defense

Packaging does much more than wrap the product. In the case of biscuits, it serves multiple vital purposes:

Function	Description
Protection	Guards biscuits from breakage, moisture, insects, and dust during transport and storage
Preservation	Maintains texture and prevents air/moisture contamination
Information	Shows manufacturing & expiry dates, ingredients, allergen info, and batch numbers
Marketing	Increases product appeal with brand logos, vibrant designs, and mascots
Convenience	Enables easy handling, resealing, or sharing (e.g., mini-packs, tear strips)

At **Independence Biscuit Company**, students observed:

- **Foil laminates** used for airtight seals
 - **Multi-layer plastic wrappers** that prevent moisture entry
 - **Tamper-proof packaging** with batch tracking for traceability
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Packaging for Sustainability

Modern packaging isn't just about safety—it must also be **environmentally responsible**. Over-reliance on plastic has led to global waste problems. That's why companies now aim for:

- **Biodegradable films**
- **Paper trays**
- **Recyclable outer boxes**
- **Ink-saving, minimal designs**

These innovations reduce pollution and appeal to eco-conscious consumers. Independence Company is also shifting to such solutions, which students got to explore during the visit.

In Summary

The importance of food preservation and packaging cannot be overstated—especially in the biscuit industry. It is a **scientific, operational, and consumer-centered practice** that protects the product, ensures food safety, builds brand trust, and supports sustainability goals.

For students, this was not just a concept but a **live experience**—watching how every biscuit is carefully handled, preserved, and packed before it reaches homes, shops, and shelves across the country.

OVERVIEW OF THE 3-DAY ACTIVITY

The educational training program at **Independence Biscuit Company** spanned over three enriching days, from **05/07/2025 to 07/07/2025**, during which students explored the **end-to-end lifecycle** of biscuit production—from raw materials to sealed packaging. Each day had a specific theme and learning outcome.

“Participant equipped with factory-standard PPE (Personal Protective Equipment) before entering production area.”



DAY 1: Introduction to Preservation Techniques in Biscuit Production

1. Welcome and Orientation Session

The first day began with a formal **welcome session** at the company’s conference hall. Participants were briefed about:

- The agenda for all 3 days
- Factory rules and safety protocols
- Objectives of the activity
- A short **corporate presentation** about the company’s history and values

The HR and Production Heads emphasized the importance of hygiene, industrial behavior, and observation. Each participant was provided with safety gear including **head caps, gloves, and shoe covers** before being taken on the facility tour.

2. Introduction to Food Preservation Science

“Understanding the cooling and drying process for baked biscuits on conveyor belts.”



Students attended a technical session on:

- The **science of food spoilage**
- Basic microbiology of **bacterial and fungal growth in food**
- Real reasons why biscuits lose their crispness over time
- Factors like **water activity (aw)**, air exposure, fat oxidation, and packaging leakage

The session was interactive, with the trainer showing how a **single batch of improperly stored biscuits** can go stale or moldy within a few days, even without obvious signs initially.

3. Factory Tour: Preservation Practices in Action

Students were taken on a **guided tour** through various sections of the factory:

- **Raw material storage** with clean, temperature-controlled bins
- **Ingredient weighing stations** where precise measurement helps consistency
- **Mixing area** where dough is prepared and flavor additives are added
- **Baking conveyor oven** where biscuits are baked evenly at controlled heat

After baking, the biscuits entered the **cooling tunnels**, where students learned about:

- How rapid cooling prevents residual baking
- How **slow or uneven cooling** can lead to biscuit softening
- Importance of **relative humidity control**

4. Focus on Biscuit-Specific Preservation Techniques

The real highlight of Day 1 was the introduction to **preservation methods** used after baking:

- **Nitrogen Flushing Machines:** These flush out oxygen and fill packets with nitrogen to extend shelf life by preventing oxidation.
- **Dehumidifier Units:** Installed in the packaging zone to keep the environment dry.
- **Moisture Monitoring Tools:** Staff demonstrated how biscuit moisture content is measured before sealing.

Students also observed use of **metal detectors**, which remove foreign particles (like screws or chips) from the conveyor before packaging.

5. Real-Time Demonstration: Spoiled vs. Preserved Batches

To make learning more practical, participants were shown:

- One set of biscuits kept in a humid room: soft, tasteless, and unsafe.
- Another set properly flushed with nitrogen: crisp, fresh-smelling, and edible even after 3 months.

The **side-by-side tasting and visual comparison** helped students understand how invisible factors like air and humidity can affect food quality.

6. Student Interactions & Q&A with Quality Team

The day ended with a round-table discussion with the **QA (Quality Assurance)** department. Students asked questions like:

- How long do your biscuits last?
- What preservatives do you use?
- Are chemical preservatives bad?
- What are natural alternatives to preservatives?

The team explained that Independence uses **minimal, approved preservatives** and relies more on **physical preservation** methods (temperature control, nitrogen flushing, moisture-proof packaging).

They also emphasized **cleanroom practices**, regular microbiological testing, and **batch recall protocols** in case of contamination.

Day 1 Takeaways

By the end of the first day, students had gained:

- A clear understanding of **what food preservation means**

- Real examples of how biscuits are preserved without harmful chemicals
- Hands-on exposure to tools like **humidity meters, nitrogen injectors**, and **cooling conveyors**
- Knowledge of why **safety and shelf life** are critical in biscuit production

The session helped turn abstract textbook terms into **real-world industrial knowledge**.

DAY 2: PACKAGING PROCESS AND DEMONSTRATION

After learning about **preservation techniques** on Day 1, the second day of the program focused on one of the most essential components of food safety and consumer experience—**packaging**. Packaging is not only about enclosing a product; it's a scientific, visual, and mechanical process that protects, promotes, and preserves.

The day was divided into 4 key segments:

- Understanding the principles of food packaging
- Learning about materials and machines used
- Watching live packaging demonstrations
- Interacting with the design and quality assurance team

1. Classroom Session: Introduction to Food Packaging Science

The session began with a multimedia presentation by the Packaging Head of the factory. Students were introduced to:

- The **roles and functions** of packaging (protection, communication, marketing, convenience)
- Types of packaging **specific to biscuits**:
 - Primary packaging (wrappers)
 - Secondary packaging (boxes/cartons)
 - Tertiary packaging (bulk crates/pallets)
- Packaging hazards: **moisture leakage, sealing failures, barcode errors**
- Importance of **barrier properties**—oxygen, light, and vapor resistance

The trainer emphasized how **packaging ensures the biscuit reaches the customer in the same condition it left the oven.**

2. Packaging Materials: A Deep Dive

“Observation of final packaging and storage of finished biscuit cartons.”



Students were shown and allowed to handle different **packaging materials** used in the factory, such as:

- **Polypropylene (PP) films** – lightweight, flexible, moisture-proof
- **Foil-laminated films** – used for high-end or export biscuits
- **Corrugated boxes** – for bulk transport
- **Food-grade ink** – used for expiry/batch coding on wrappers

There was also a discussion on:

- **Eco-friendly alternatives** like recyclable mono-materials
- Limitations of biodegradable films (cost, limited barrier strength)
- **Ink migration** and how packaging inks are made food-safe

Samples were passed around for students to examine the texture, print quality, and sealing area.

3. Factory Walkthrough: Live Packaging Demonstration

The highlight of Day 2 was a **step-by-step walk-through of the live packaging lines**.

Conveyor Belt Observation:

Biscuits were seen moving from the cooling tunnel to the sorting trays. Workers inspected and rejected:

- Cracked or burnt biscuits
- Underbaked or odd-shaped units

Wrapping Line:

- Biscuits entered **horizontal flow-wrap machines** that cut, wrap, and seal hundreds of packets per minute.
- Students observed the **crimp-seal and heat-seal technology** that keeps packets airtight.
- Faulty packets were auto-rejected using **weight sensors and camera-based detection**.

Secondary and Tertiary Packaging:

- Individual packets were stacked and inserted into **printed cartons**.
- Cartons were sealed, coded, and sent to the dispatch zone.
- Each carton had a unique **QR code, batch ID, MRP, and expiry date**.

This demonstration gave students a chance to **observe automation in action**, and how precise every movement must be for consistency and efficiency.

4. Discussion with the Design & Quality Team

Post the walk-through, students engaged in a Q&A session with:

- **Packaging Designers**
- **Printing Supervisors**
- **Quality Assurance Inspectors**

Key learning points included:

- **How packaging affects consumer trust:** Clean, sealed, well-labeled packages build confidence.
- **Why printing clarity and compliance** (FSSAI logos, vegetarian symbols, ingredient list) matter legally.
- Challenges in designing **tamper-evident** and **child-safe packaging**
- Trends in **sustainable packaging innovation** (e.g., plastic-free pouches)

Students also learned how machines are calibrated:

- For **weight accuracy**
- **Roll tension** for wrapper film
- **Temperature control** in sealing jaws

One exciting live demo showed how a small **pin-sized hole** in a biscuit pack can lead to complete spoilage within days.

Activity: Design Your Own Packaging

To end the day on a creative note, students were split into teams and given an activity:

“Design a biscuit wrapper for a new product.”

They were asked to consider:

- Product name
- Logo
- Color scheme

- Material
- Safety info
- Eco-friendliness

Each group presented their ideas, and the design team gave feedback on **practicality, marketing, and material suitability**.

Day 2 Takeaways

- Packaging is a **blend of engineering, science, marketing, and sustainability**.
 - Students understood the **importance of barrier properties**, sealing quality, and design clarity.
 - Real-time exposure to machines made abstract textbook concepts **come alive**.
 - It inspired students to consider careers in **packaging technology, graphic design for FMCG, or industrial QA**.
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DAY 3: HANDS-ON ACTIVITY & QUALITY CONTROL IN BISCUIT PRODUCTION

The third and final day of the industrial visit at **Independence Biscuit Company** was the most engaging and participatory of the entire experience. After two days of observing preservation and packaging techniques, Day 3 was dedicated to **hands-on learning, quality testing, and experiencing real-time industrial roles.**

It gave students the chance to:

- Apply their theoretical knowledge
- Operate under supervision
- Understand the importance of food safety regulations
- Learn from QA (Quality Assurance) and QC (Quality Control) processes

1. Recap and Team Formation

The day began with a **brief recap session** of Days 1 and 2. Students shared reflections, and trainers highlighted key lessons to remember. Then, the participants were split into **small functional teams** to rotate across different activity stations.

Each team had a mentor or floor supervisor assigned to guide them through:

- Packaging trials
- Weight checking
- Sealing tests
- Quality inspections

This “mini-internship” style layout created an immersive experience that made students feel like active contributors to the factory’s workflow.

2. Hands-On Activity Stations

Station 1: Manual Packaging

Participants packed biscuits into small pouches manually. This helped them understand:

- The **importance of speed and precision**
- Challenges in ensuring equal quantity and proper sealing
- Why human errors are still a risk in semi-automated lines

Station 2: Machine Monitoring

Each team observed and partially controlled a **semi-automated packaging machine**. They learned how:

- Film rolls are loaded and aligned
- Temperature settings affect the seal
- Timing, film feed, and sensor calibration keep the process accurate

Supervisors demonstrated how a **minor delay in sealing** can lead to improper packs or rejected batches.

Station 3: Labeling and Coding

Students entered **batch codes and expiry dates** using programmable print units. They saw:

- How inkjet and laser coders print on high-speed belts
- Why **clear, permanent coding** is essential for traceability
- What batch numbering means in case of a product recall

3. Quality Control Lab Tour and Testing

“Freshly baked biscuits moving along the inspection track for quality checks.”



The most scientific part of Day 3 was the **QC laboratory tour**, where students:

- Learned how samples from each production batch are tested
- Saw real **moisture analyzers, texture testers, and pH measurement tools**
- Observed microbial plating tests used to detect contamination

QA staff explained how even a single contaminated batch can damage brand reputation and pose **health risks to consumers**.

Key QC Checks Explained:

Check Type	Purpose
Moisture Content	Ensures crispness and shelf life
Weight Accuracy	Avoids customer complaints / legal fines
Seal Integrity	Prevents air/moisture entry
pH Level	Detects fermentation or spoilage
Visual Defects	Filters broken, burnt, or misprinted packs

Students were allowed to **test real biscuit samples**, record data, and interpret results.

4. Industry Insights & Career Guidance

A special session was conducted with:

- **Production Heads**
- **R&D Officers**
- **Food Safety Auditors**

They shared insights on:

- How new flavors and biscuit types are researched
- What skills are needed for roles in QA, Production, or Packaging
- How government agencies like **FSSAI** and **ISO bodies** monitor standards

They also advised students on:

- Certification courses in **Food Technology, HACCP, and GMP**
 - Internship opportunities and resume-building tips
 - Soft skills like teamwork, documentation, and analytical thinking
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Certificate Distribution & Closing Ceremony

At the end of Day 3:

- Participants were awarded **certificates of participation**
- Group photos were taken at the company premises
- Feedback was collected from students and faculty

Many students expressed how the experience had:

- Changed their view of “simple” foods like biscuits
 - Inspired them to consider **industrial roles**
 - Shown how even behind a ₹5 product lies deep science and process
-

Day 3 Takeaways

- Students gained **real-time operational experience**
- They understood how **quality testing** directly affects safety and brand trust
- The hands-on sessions boosted their **confidence and technical clarity**
- They were encouraged to pursue **careers and innovations** in the food industry

This day concluded the training on a **motivational and insightful note**, combining practical learning with inspiration for future possibilities.

LEARNINGS AND TAKEAWAYS

The 3-day industrial activity at *Independence Biscuit Company* offered a **rich blend of observation, hands-on experience, and technical understanding**. It transformed theoretical classroom knowledge into meaningful, real-world exposure.

Here are the major takeaways:

Technical Learnings:

- The **importance of temperature, humidity, and hygiene** in biscuit preservation.
- Exposure to **automated machines** such as flow wrappers, nitrogen flushers, and date coders.
- Understanding the **role of packaging in safety, branding, and shelf life**.
- How **moisture testing, weight checks, and microbial tests** ensure food quality.

Educational Outcomes:

- Students learned how even basic food products require **advanced technology and strict standards**.
- Topics like **FSSAI compliance, batch tracking, sealing systems, and barcoding** were demystified.
- They realized the significance of **process timing, coordination, and human-machine interaction** in mass production.

Personal and Professional Growth:

- Improved confidence in speaking to professionals and asking technical questions.
- Boosted interest in **careers in food processing, packaging technology, and industrial QA**.
- New appreciation for **teamwork, cleanliness, and responsibility** in production environments.

This experience left students more **industry-aware, self-assured, and curious about innovations** in food science.

CHALLENGES FACED

Though the experience was highly informative, students encountered certain **real-world industrial challenges**, which made the visit even more valuable:

1. Understanding Complex Machines

- High-speed packaging lines were difficult to follow at first due to their speed and technicality.
- Students needed support to understand how machines operate in synchronization.

2. Hygiene Protocols

- Strict hygiene rules meant students had to wear full safety gear and avoid touching anything without permission, limiting initial interactivity.

3. Technical Jargon

- Terms like *crimp sealing*, *roll tension*, and *product recall protocols* were unfamiliar, requiring live explanations from experts.

4. Limited Interaction Time

- Factory work is time-bound, and employees are on fixed schedules, so student interaction windows were short but intensive.

Despite these challenges, students adapted quickly and learned how **real industries are fast-paced, efficient, and highly organized**.

INDUSTRY INSIGHTS

The visit opened students' eyes to the **breadth and depth of the food industry**, particularly the biscuit sector.

Here are some key insights:

The Biscuit Industry is a High-Volume Business

- Even small biscuit plants produce **lakhs of units per day**.
- A minor delay or quality issue can result in **huge losses or market complaints**.

Safety and Trust are Non-Negotiable

- Consumers may not see the back-end effort, but factories ensure every step maintains **strict safety and hygiene**.

Automation and AI are Rising

- Companies are shifting toward **sensor-based quality checks and automated reject systems**.
- Even printing, coding, and sealing are getting smarter.

Sustainability is the Future

- There is a **clear move toward biodegradable, recyclable, and eco-friendly packaging**.
- Students realized that **they could contribute innovative ideas** in this direction through research and project work.

CONCLUSION

The 3-day activity at **Independence Biscuit Company, Mysuru**, was more than an industrial visit. It was a **gateway into the real world of food manufacturing, safety, and packaging technology**. Students saw firsthand how something as common as a biscuit is the result of **precision, care, planning, and science**.

From dough mixing to baking, from preservation to sealing, every step is governed by **strict quality measures, automation, and innovation**. The interaction with professionals helped bridge the gap between **classroom learning and industrial reality**.

Students now understand:

- The importance of **food engineering**
- The potential of careers in **packaging, QA, and food tech**
- The responsibility the food industry carries in feeding people safely

This activity ignited a spark in many participants to **pursue food-related innovation**, to think sustainably, and to value the complexity behind even the simplest packet of biscuits.

The visit concluded with a renewed respect for the industry, a clearer academic direction, and a strong sense of purpose.

ANNEXURE

The following images were taken during the 3-day food preservation and packaging activity at *Independence Biscuit Company*, Mysuru. They provide a visual representation of the learning experiences, factory setup, safety protocols, packaging processes, and student participation. These photographs capture key moments and stages of the biscuit production lifecycle from preparation to packing.

Image 1: Entry to Independence Biscuit Factory



Image 2: Stacked Corrugated Boxes for Packaging



Image 3: Live View of Biscuit Packaging Conveyor



Image 4: Students Observing Packing Line



Image 5: Inside the Biscuit Manufacturing Unit



Image 6: Products Ready for Dispatch

